

Nathan Stangler

Minneapolis, MN
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EDUCATION

Master of Science in Computer Science (Integrated Program)
University of Minnesota-Twin Cities

Expected May 2027
Minneapolis, MN

Bachelor of Science
University of Minnesota-Twin Cities
Majors: Computer Science, Data Science; Minor: Statistics
GPA: 3.973/4.00

Expected May 2026
Minneapolis, MN

PROFESSIONAL EXPERIENCE

ImageTrend

May 2025-August 2025

Software Developer Intern

Eagan, MN

- Designed and implemented a full-stack geocoding data visibility and control application with a Windows service API using C# and .NET Core.
- Created an interface for managing geocoding data sources and viewing logs using React.
- Refactored legacy code to improve performance and integrate latest features.

Best Buy

July 2021-April 2023

Sales Advisor

Maple Grove, MN

- Trained new employees in technical aspects of mobile device activation and setup.
- Provided technical support for customers by using critical thinking skills to ease concerns.

RESEARCH EXPERIENCE

University of Minnesota, Knowledge Computing Lab

May 2025-Present

Undergraduate Researcher (Mentor: JangHyeon Lee, Advisor: Yao-Yi Chiang)

Minneapolis, MN

- Led development of SolarCLIP, a domain-adapted vision-language model for solar physics by selectively fine-tuning CLIP's projection heads and targeted transformer layers.
- Designed and ran a full machine learning experimentation pipeline including model training, evaluation tooling, data preprocessing, and embedding-space analysis.

TEACHING EXPERIENCE

University of Minnesota, Department of Science & Engineering

January 2026-Present

Undergraduate Teaching Assistant (CSCI 5541: Natural Language Processing)

Minneapolis, MN

- Supporting instruction for a graduate-level NLP course covering word embeddings, transformers, and large language models.
- Mentoring students through semester-long projects involving dataset collection, annotation, model design, training, and evaluation methodologies.
- Providing detailed grading and technical feedback on assignments and projects.

SELECTED PROJECTS

VulnRL

October 2023-Present

- Designing a compiler-verified reinforcement learning framework for C/C++ vulnerability detection using compiler diagnostics and analyzer feedback as reward signals.
- Implementing a pipeline combining LoRA-based parameter-efficient fine-tuning and GRPO reinforcement learning on large code LLMs to enable CWE-level vulnerability classification.

Truncate

Summer 2023

- Built a full-stack URL shortener and analytics platform using React, TypeScript, Go (Fiber), and PostgreSQL.
- Implemented detailed analytics, including redirect tracking, geolocation, device, and browser statistics.
- Secured user data with JWT authentication and role-based access control.

Improved Scoreboard

Summer 2024

- Designed and developed a scalable, object oriented, and event driven Minecraft network packet-based scoreboard plugin compatible with multiple game versions.
- Integrated with third-party applications through the creation of a portable scoreboard management API.

Daily Rewards

Fall 2020

- Developed and commercially launched a fully configurable daily rewards system for Garry's Mod, generating 100+ sales on Gmodstore.
- Integrated MySQL and SQLite database support, with seamless interaction from a custom designed user interface using client-server communication through networking protocols.

ACTIVITIES

Member, University of Minnesota Robotics Club

September 2023-May 2024

- Contributed to the creation and design of an autonomous robot for the NASA Lunabotics competition using NVIDIA Isaac ROS libraries.
- Collaborated with club members to develop and debug a heightmap conveyor belt detector and a vision-based position estimator.

AWARDS AND HONORS

Dean's List, University of Minnesota, Minneapolis, MN, 2023-Present

SKILLS

Languages: Java, Python, C, C#, SQL, JavaScript/TypeScript, HTML, CSS, JSON

Libraries: PyTorch, NumPy, Pandas, Scikit-Learn, React, Material UI, Jest

Tools: Git, GitHub, HuggingFace, Weights & Biases, PostgreSQL